

The Kelly Criterion: How Much to Bet?

Srinjena Das

DMS DAY 2024

A **gambling** game can be either

1. discrete, or,
2. continuous.

Definition (Favorable game)

A game in which there exists a strategy such that

$$P(\lim_{n \rightarrow \infty} X_n = \infty) > 0,$$

where $X_n \equiv$ capital after n trials.

Discrete Gambling Game

A biased coin is tossed, with our win probability being $p > \frac{1}{2}$, and hence, $q = 1 - p < \frac{1}{2}$ is our probability of loss.

- $X_0 \equiv$ Initial capital.
- Our opponent is infinitely rich.
- The players of the game place **even money bets** on repeatedly independent trials of tossing the biased coin.

Maximizing the Expected Value $E(X_n)$

Let B_k be the amount to be bet on the k – th trial.

Consider

$$T_k = \begin{cases} 1 & \text{if } k\text{-th trial is a win} \\ -1 & \text{if } k\text{-th trial is a loss.} \end{cases}$$

Then,

$$X_k = X_{k-1} + T_k B_k, \quad k = 1, 2, 3, \dots,$$

$$\text{i.e. } X_n = X_0 + \sum_{k=1}^n T_k B_k, \quad \forall n \geq 1,$$

$$\text{and hence, } E(X_n) = X_0 + \sum_{k=1}^n E(T_k B_k).$$

Now, $E(T_k) = 1 \times p + (-1) \times q = p - q$, and as T_k and B_k are independent random variables, it follows that

$$E(X_n) = X_0 + \sum_{k=1}^n (p - q)E(B_k).$$

Maximizing the Expected Value $E(X_n)$

Thus, to maximize expected gain, the strategy that should be followed is

$$B_1 = X_0, B_2 = 2X_0, \dots$$

The probability of ruin at the n -th trial is given by $1 - p^n$, where $\frac{1}{2} < p < 1$, that is, as $n \rightarrow \infty$, $1 - p^n \rightarrow 1$. Hence, the ruin is sure!

Minimizing the Probability of Ruin

As in the case of gambler's ruin, if $X_0 = i$, and,

$$B_j = \begin{cases} 1 & \text{if } 0 < X_j < N \\ 0 & \text{if } X_{j-1} = 0, N \end{cases}$$

then, the probability of ruin $\tilde{q}$ is given by

$$\tilde{q} = \frac{\left(\frac{q}{p}\right)^i - \left(\frac{q}{p}\right)^N}{1 - \left(\frac{q}{p}\right)^N}. \quad (1)$$

So, if the amount bet on each trial is $\frac{1}{r}$ units instead of 1 unit, where $r \in \mathbb{N}$, then, it is equivalent of starting the game with initial capital ir . Considering $q(r)$ to be the probability of ruin in such case, from (1) we have

$$q(r) = \frac{\left(\frac{q}{p}\right)^{ir} - \left(\frac{q}{p}\right)^{Nr}}{1 - \left(\frac{q}{p}\right)^{Nr}}. \quad (2)$$

Since, $p > q$, taking $\frac{1}{r} \rightarrow 0$, that is, $r \rightarrow \infty$ in equation (2), we have $q(r) \rightarrow 0$. Thus, by betting **sufficiently small** at each trial, the probability of ruin can be made arbitrary small. But beware! This **reduces the expected gain**.

An Intermediate Strategy: The Kelly Criterion

In the coin tossing game, the following two things remains constant:

- gambling probabilities.
- even pay-off situation.

It seems plausible that an “optimal” strategy will involve always wagering the same fraction f (assuming that our capital is infinitely divisible) of the current bankroll in every trial. We call this strategy the “fixed fractional” strategy.

So, at the i – th stage, $B_i = fX_{i-1}$, where $0 \leq f \leq 1$, $\forall i \geq 1$.
Henceforth, till the n – th trial, if

$S_n \equiv$ no. of successes

$F_n \equiv$ no. of failures,

then our capital after n trials is given by

$$X_n = X_0(1 + f)^{S_n}(1 - f)^{F_n}, \quad (3)$$

where $S_n + F_n = n$.

Clearly, $\mathbf{P}(X_n = 0) = 0$, so ruin shall be reinterpreted to mean that for arbitrary small $\epsilon > 0$, $\lim_{n \rightarrow \infty} [\mathbf{P}(X_n \leq \epsilon)] = 1$.

Now, from (3),

$$\log \left(\frac{X_n}{X_0} \right)^{1/n} = \frac{S_n}{n} \log(1 + f) + \frac{F_n}{n} \log(1 - f). \quad (4)$$

From the fact that

$$e^{n \log \left(\frac{X_n}{X_0} \right)^{1/n}} = \frac{X_n}{X_0},$$

it follows that the expression in eq. (4) represents the exponential rate of increase per trial, also called the **growth rate co-efficient**.

Let us define

$$\begin{aligned} G(f) &:= E \left[\log \left(\frac{X_n}{X_0} \right)^{1/n} \right] = E \left[\frac{S_n}{n} \log(1+f) + \frac{F_n}{n} \log(1-f) \right] \\ &= p \log(1+f) + q \log(1-f). \end{aligned}$$

The Kelly strategy involves maximizing the function $G(f)$, i.e. *maximizing the expected value of the growth rate co-efficient*.

It can be shown that the function $G(f)$ has a global maximum at $f = p - q$, which we denote as f^* , where

$G(f^*) = \log 2 + p \log p + q \log q > 0$. Also, there exists unique f_c such that $f^* < f_c < 1$ and $G(f_c) = 0$.

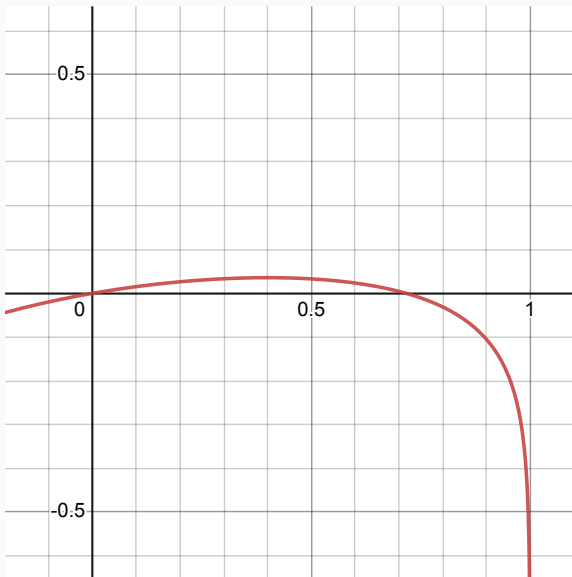


Figure 1: The graph of $G(f)$ with $p = 0.7$.

Why is f^* the Optimal Fraction in the Class of All Fixed Fractional Strategies?

Theorem

1. If $G(f) > 0$, then, $\lim_{n \rightarrow \infty} X_n = \infty$ a.s., that is, for each $M > 0$,
 $P(\liminf_{n \rightarrow \infty} X_n > M) = 1$.
2. If $G(f) < 0$, then, $\lim_{n \rightarrow \infty} X_n = 0$ a.s., that is, for each $\epsilon > 0$,
 $P(\limsup_{n \rightarrow \infty} X_n < \epsilon) = 1$.
3. If $G(f) = 0$, then $\limsup_{n \rightarrow \infty} X_n = \infty$ a.s., and, $\liminf_{n \rightarrow \infty} X_n = 0$ a.s.
4. If $G(f_1) > G(f_2)$, then, $\lim_{n \rightarrow \infty} \frac{X_n(f_1)}{X_n(f_2)} = \infty$ a.s.

The Kelly Strategy Dominates over "All" Strategies

Consider a series of independent trials $\{X_n\}$ in which

$Q_i \equiv$ Per unit return on the i -th bet.

Note that,

$$\frac{X_n}{X_0} = \prod_{i=1}^n \left(\frac{X_i}{X_{i-1}} \right), \text{ and thus, } E \left[\log \left(\frac{X_n}{X_0} \right) \right] = \sum_{i=1}^n E \left[\log \left(\frac{X_i}{X_{i-1}} \right) \right].$$

Now, for each i , we have $X_i = X_{i-1} + B_i Q_i$, whence

$$\frac{X_i}{X_{i-1}} = 1 + \left(\frac{B_i}{X_{i-1}} \right) Q_i.$$

Considering $F_i = \frac{B_i}{X_{i-1}}$, we have that

$$E \left[\log \left(\frac{X_n}{X_0} \right) \right] = \sum_{i=1}^n E[\log(1 + F_i Q_i)],$$

where F_i and Q_i are independent.

Theorem

If for each i , there is an f_i , $0 < f_i < 1$ such that $E[\log(1 + F_i Q_i)]$ is defined and positive, then for each i , there is a number f_i^ such that $E[\log(1 + F_i Q_i)]$ attains its unique maximum for $F_i = f_i^*$ a.s. (To avoid trivialities we assume $Q_i \neq 0$ for each i .)*

Continuous Approximation of Expected Time to Reach A Goal

The optimal growth strategy also **asymptotically** minimizes the expected time to reach a goal, that is if

- $m(C) \equiv$ Greatest lower bound over all strategies for the expected time to reach C , and,
- $t^*(C) \equiv$ Expected time using Kelly strategy,

then,

$$\lim_{C \rightarrow \infty} \frac{t^*(C)}{m(C)} = 1.$$

The continuous approximation to the expected number of trials to reach the goal $C > X_0 = 1$ is

$$n(C, f) = \frac{\log C}{G(f)},$$

where f is any fixed strategy. Since, $G(f)$ has a unique maximum at f^* , $n(C, f)$ has a unique minimum at $f = f^*$.

Continuous Gambling Game

In Accordance with Investing in Stock Market

Consider the random variable R where

$R \equiv$ Return per unit from investment made in stocks,

in which $P(R = \mu + \sigma) = P(R = \mu - \sigma) = 0.5$, where, $\mu = E[R]$ & $\sigma^2 = Var(R)$.

Also, assuming

- $V_0 \equiv$ Initial capital,
- $r \equiv$ Rate of return of capital invested elsewhere,
- $V(f) \equiv$ Capital after receiving returns from all the investments,

we obtain

$$V(f) = V_0(1 + (1 - f)r + fR).$$

Dividing our time into n equal intervals, we get a sequence of random variables $\{V_n(f)\}$ and following the lines of our discrete case, we have

$$G_n(f) = \sum_{i=1}^n E \left[\log \left(1 + (1-f) \frac{r}{n} + f R_i \right) \right].$$

Expanding $G_n(f)$ in power series around $f = 0$, we obtain

$$G_n(f) = r + f(\mu - r) - \frac{\sigma^2 f^2}{2} + O(n^{-k}).$$

Taking $n \rightarrow \infty$, we get

$$E \left[\frac{V(f)}{V_0} \right] = G_\infty(f) = r + f(\mu - r) - \frac{\sigma^2 f^2}{2},$$

which is our required function that we need to maximize, and thus maximizing $G_\infty(f)$, we obtain our required Kelly fraction as

$$f^* = \frac{\mu - r}{\sigma^2}.$$

S&P 500 Index (1926-1984): The Case with $f^* > 1$

For the time period of 59 years (1926-1984), the mean μ and standard deviation σ of the returns were given by

$$\mu = .058 \quad \sigma = .216.$$

Therefore, $f^* = \frac{\mu}{\sigma^2} \approx 1.24$, that is, we should be willing to bet **124%** of our initial resources!

Because of the significant downside tail-risk of equity markets, investors might choose to reduce to **half-Kelly**.

- [1] Thorp, Edward O., The Kelly criterion in blackjack, sports betting and the stock market. Edward O. Thorp & Associates, Newport Beach, 1997.
- [2] Breiman, Leo and others, Optimal gambling systems for favorable games, 1961.
- [3] Feller, William, An introduction to probability theory and its applications, Volume 2, John Wiley & Sons, 1991.
- [4] Rotando, Louis M and Thorp, Edward O, The Kelly criterion and the stock market, The American Mathematical Monthly, Volume 99, Number 10, 922-931, Taylor & Francis, 1992.